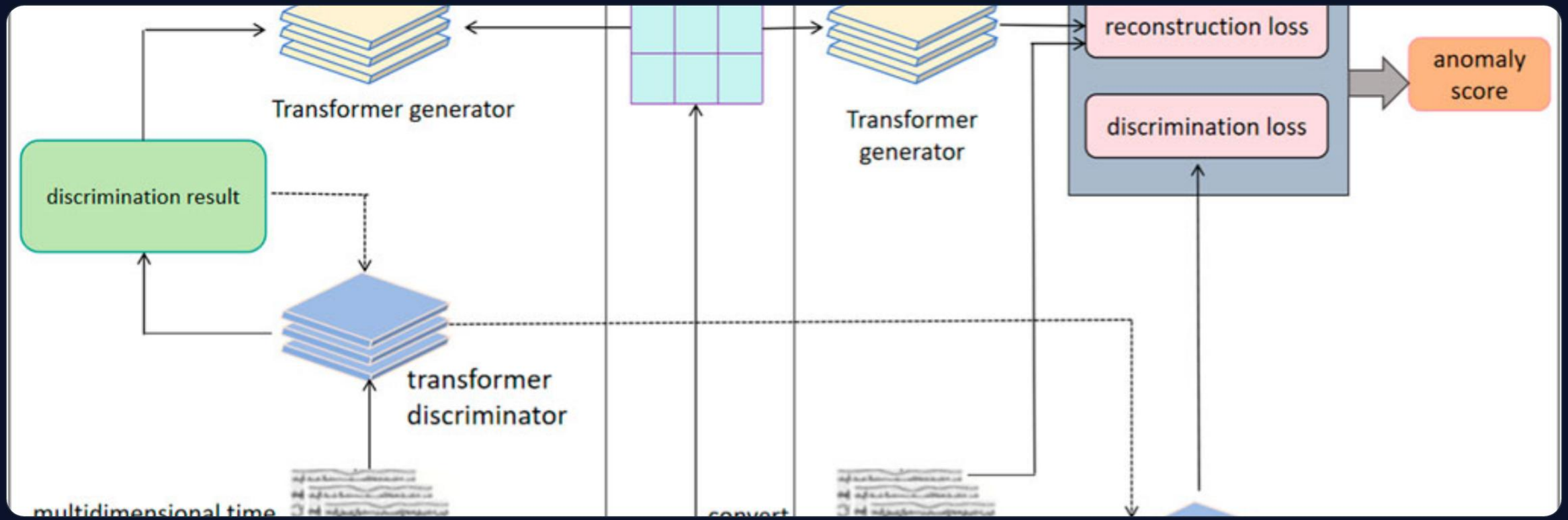
CAPSTONE PROJECTS | SECOND-YEAR EEE

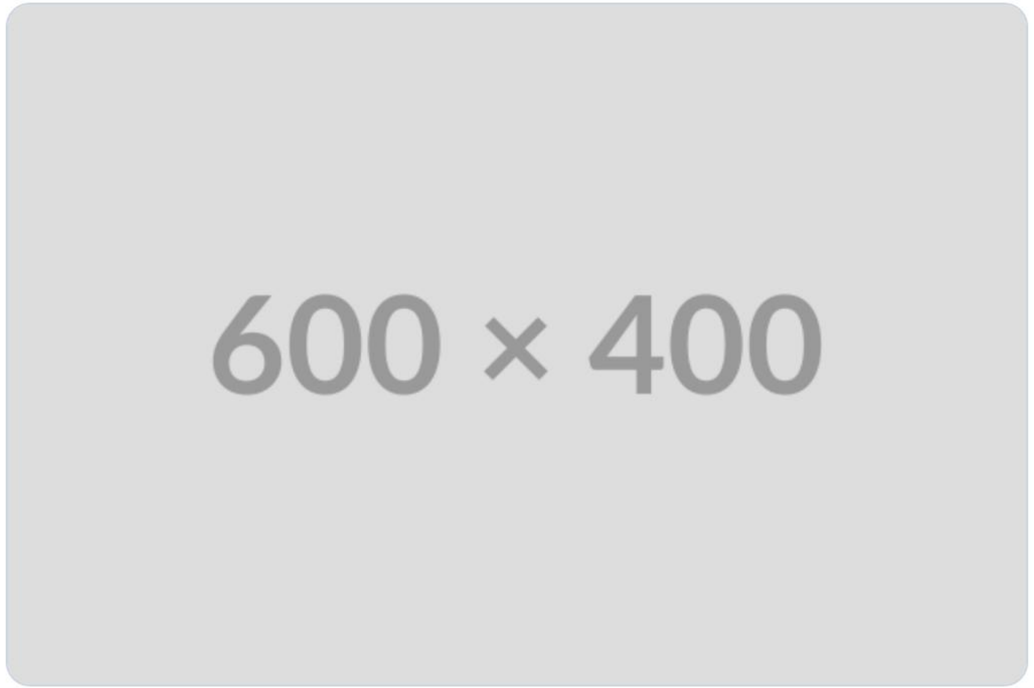
CAPSTONE – 6: AI-Based Energy Theft / Abnormal Consumption Pattern Detection

Machine Learning Framework for Automated Screening & Anomaly Scoring in Smart Grids



1. SUB-TITLE: PROJECT QUESTION

What Do You Have To Do? & System Definition



Project Definition

Develop an ML-based educational system that identifies unusual electricity-consumption patterns from meter readings or public datasets to flag anomalies for investigation.

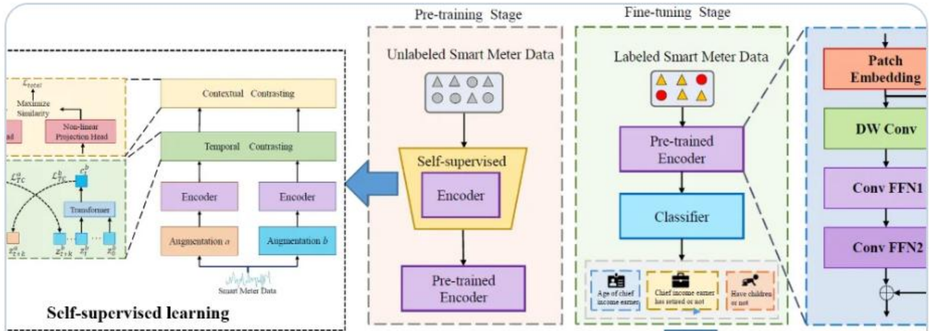
Core Objective & Scope

Build clean baseline consumption models and rank flag intervals using anomaly scores without claiming that an anomaly alone proves electricity theft.

2. SUB-TITLE: REQUIRED AI SKILLS

Core Competencies & Technical Matrix

AI Skill	Requirement Purpose
Python Programming	Data processing & script development
Data Preprocessing & EDA	Missing values, outliers, time-of-day & peak usage
Feature Engineering	Historical averages & consumption delta features
Anomaly Models	Isolation Forest, Local Outlier Factor, Random Forest
Model Evaluation & Viz	Precision, recall, F1-score & Streamlit timeline



Performance

Method	Task300			Task310			Task410		
	Acc.	Rec.	F1	Acc.	Rec.	F1	Acc.	Rec.	F1
SimCLR[29]	70.70	68.52	69.23	69.42	67.85	68.54	62.40	60.24	61.17
CPC[17]	72.86	71.18	71.85	73.84	72.53	73.12	63.91	62.81	63.24
ClsTran	72.79	71.76	72.18	74.75	73.62	74.03	63.26	62.48	62.83
TimeMAE[38]	74.11	73.52	73.78	76.99	76.24	76.53	65.49	64.82	65.14
ResNet1d_wang[21]	68.15	67.21	67.53	71.40	70.54	70.87	61.90	60.53	61.12
Informer[24]	68.99	67.83	68.25	72.59	71.52	72.08	61.33	60.27	60.73
Autoformer[25]	70.19	69.48	69.82	74.53	73.85	74.18	62.85	61.92	62.27
Timesnet [16]	67.64	66.53	66.92	72.92	71.84	72.25	61.01	60.23	60.58
ModernTCN[15]	71.65	70.63	70.94	74.79	73.93	74.25	62.77	61.82	62.23
Ours	**75.76**	**75.18**	**75.52**	**78.04**	**77.53**	**77.84**	**65.81**	**64.93**	**65.32**

3. SUB-TITLE: TOOLS & TECHNOLOGIES

Software Environment & Data Pipeline

Environment

Python, Jupyter, Colab for reproducible exploratory data analysis.

ML Stack

Pandas, NumPy, Scikit-learn, Matplotlib, and Seaborn for models & graphs.

Dashboard UI

Streamlit monitoring dashboard for reviewing flagged consumer intervals.

Data Sources

Public meter datasets, simulated logs, or optional smart hardware meters.

APPLICATIONS, ADVANTAGES & DISADVANTAGES

System Feasibility & Operational Impact

Applications

Utility Grid Screening: Pre-screens millions of smart meter feeds efficiently.

Non-Technical Loss Reduction: Minimizes unbilled revenue loss for power companies.

Fault Isolation: Helps locate defective or bypass-manipulated meters.

Advantages

Non-Invasive Audit: Evaluates remotely without continuous physical inspections.

Automated Scoring: Quantifies probability using transparent scoring.

Scalable: Handles large multi-household interval datasets easily.

Disadvantages

False Positives: Sudden genuine consumer lifestyle shifts trigger alerts.

Screening Only: Cannot serve as legal proof without physical verification.

Data Sensitivity: Requires clean, uncorrupted consumption time-series data.

4. SUB-TITLE: FINAL SUBMISSION REQUIREMENTS

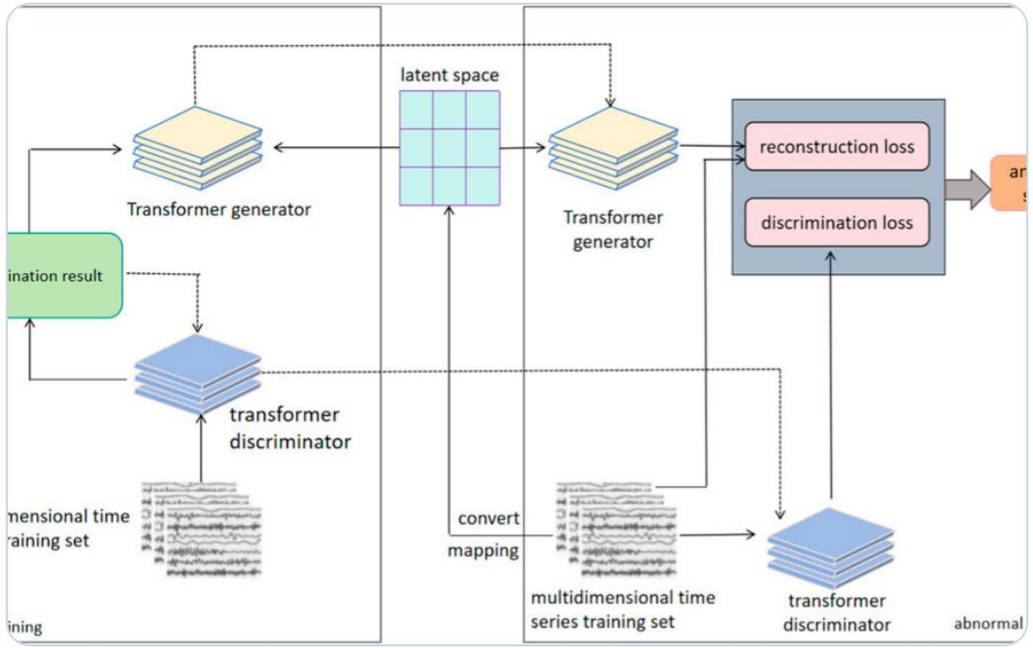
Evaluation Checkpoints & Viva Requirements

Dataset & Assumptions: Defined source data, assumptions, preprocessing, and EDA notebook.

Model & Feature Code: Complete feature engineering code alongside trained anomaly models.

Evaluation Results: Anomaly score distribution, test case analysis, and false-positive discussion.

Dashboard & Report: Interactive Streamlit dashboard, project report, and viva presentation.

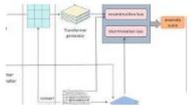


CAPSTONE PROJECT COMPLETE

AI-Based Energy Theft / Abnormal Consumption Pattern Detection

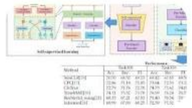
Responsible AI solution balancing automated screening efficiency with authorized investigation requirements.

Image Sources



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